

Coding & Solution

Date	30 June 2026
Team ID	xxxxxxx
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/Rohit12223/AI ML-TRACK- _INTERNSHIP
Programming Language(s)	Python 3.x, HTML5, CSS3, JavaScript
Framework(s) Used	Flask (web framework), Scikit-learn (ML), Pandas, NumPy, Matplotlib, Seaborn, Pickle

Key Features Implemented	1. HDI score prediction using Linear Regression trained on UNDP 2021 data. 2. Live HDI score estimator that updates as user types. 3. Automatic classification into Low / Medium / High / Very High HDI. 4. Dark themed interactive web UI with progress tracker. 5. Flask web app with 3 pages — home, prediction form, result page .
Pending / Incomplete Features	Country dropdown selection was removed to simplify input. Internet Users field was removed as it was not in the final dataset. Future scope includes adding more countries and deploying to cloud
Setup / Run Instructions	1. pip install numpy pandas matplotlib scikit-learn Flask seaborn. 2. Download HDI.csv from Kaggle and place in Dataset/ folder. 3. Open Training/HumDevIndex.ipynb and run all cells to generate HDI.pkl. 4. Open terminal, cd Flask, run python app.py . 5. Open browser at http://localhost:5000

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes

4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The HDI Predictor project is a complete end-to-end machine learning web application. The Jupyter notebook (HumDevIndex.ipynb) handles all training steps from data loading to model saving. The Flask backend (app.py) loads the trained HDI.pkl model and serves predictions through a clean dark-themed web interface. All code follows Python best practices with clear comments, meaningful variable names, and modular structure. The live HDI estimator on the prediction form page uses JavaScript for real-time score estimation without a server call.